

COVID-19 Data Analysis Project (Python + PostgreSQL)

Project Description

This project focuses on performing an end-to-end data analysis of the global COVID-19 pandemic using real-world data. The analysis is conducted using Python for data processing, visualization, and feature engineering, along with PostgreSQL for structured querying and insight extraction.

The dataset used in this project is sourced from the Our World in Data COVID-19 dataset available on Kaggle, which contains detailed information on COVID-19 cases, deaths, and vaccination trends across multiple countries over time.

The primary objective of this project is to clean, transform, and analyze large-scale pandemic data to identify meaningful trends and patterns. The project involves handling missing values, converting data types, and creating new features such as death rate to enhance analytical insights.

Using data visualization techniques, the project explores trends such as the growth of total cases, daily new cases, death rates, and vaccination progress for specific countries like India, as well as globally. Graphs and charts are created using Matplotlib and Seaborn to make the analysis more intuitive and visually interpretable.

In addition to Python-based analysis, the cleaned dataset is exported and stored in a PostgreSQL database. SQL queries are then used to perform deeper analysis, such as identifying top affected countries, calculating death rates, analyzing monthly trends, and evaluating vaccination progress across different regions.

This project demonstrates practical skills in data cleaning, exploratory data analysis (EDA), data visualization, and database querying, making it a strong entry-level portfolio project for data analyst roles.

Problem Statement

The COVID-19 pandemic generated massive amounts of global data, but extracting meaningful insights from this data is challenging due to its size, inconsistency, and missing values.

The problem addressed in this project is:

- How can we analyze global COVID-19 data to identify trends in cases, deaths, and vaccinations?
- How can we clean and preprocess real-world data to make it suitable for analysis?
- How can we use visualization techniques to understand patterns over time?
- How can SQL be used to extract meaningful insights from structured data?

This project aims to solve these challenges by building a complete data analysis pipeline using Python and PostgreSQL.

Tools & Technologies Used

- Python (NumPy, Pandas)
 - Data Visualization (Matplotlib, Seaborn)
 - PostgreSQL (SQL Queries)
 - Google Colab
 - Kaggle Dataset (Our World in Data COVID-19)
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Key Objectives

- Perform data cleaning and preprocessing on raw COVID-19 data
- Handle missing values and inconsistent data
- Conduct exploratory data analysis (EDA)
- Visualize trends in cases, deaths, and vaccinations
- Create new features such as death rate and growth rate
- Store processed data in PostgreSQL
- Write SQL queries to extract business insights

Expected Outcomes

- Identify top countries affected by COVID-19
- Analyze trends in daily and total cases
- Understand the relationship between cases and deaths
- Evaluate vaccination progress globally
- Generate insights using both Python and SQL